

CS383 - Software Engineering

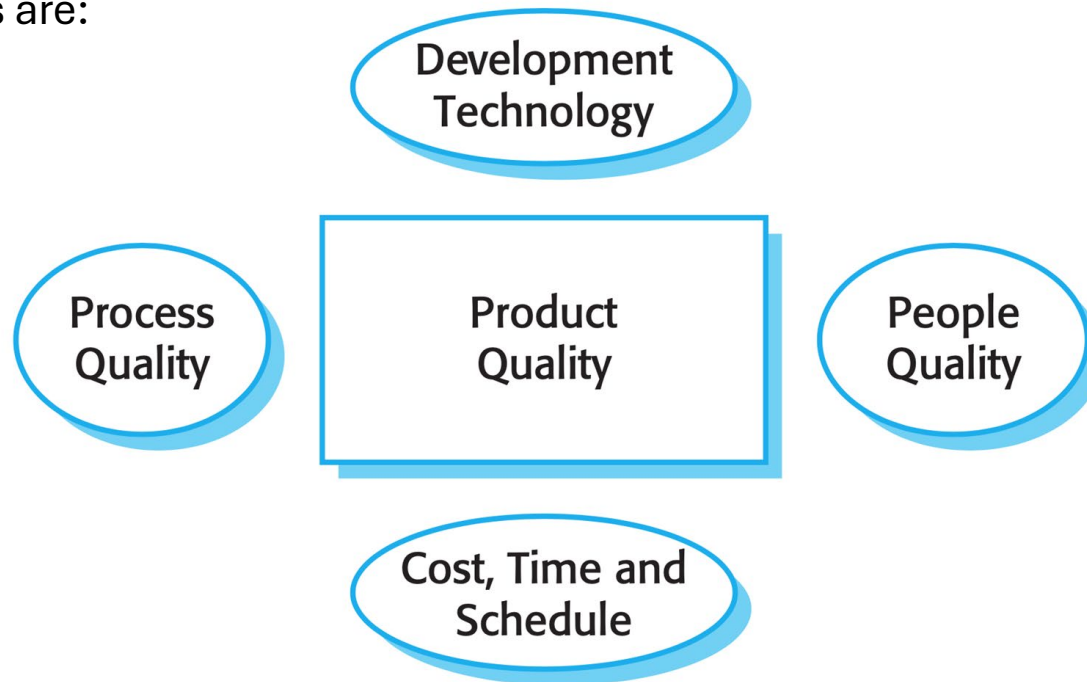
Software Process Quality

Outline

- Product quality and quality factors
- Software process attribute
- Software process improvement
 - Measurement, analysis and change
- Software process classification
- Software standards

Process and product quality

- Product quality depends on development process
 - Good process yields usually good product
- Software engineers analyze current software process for improvement changes
- Factors are:



Process and product quality (cont.)

- The quality for big projects is determined by
 - The development process
- The quality for small projects is determined by
 - Developers skills and capability
 - Development technologies
- Whatever the project size, unrealistic schedule leads to poor quality product

Process attributes

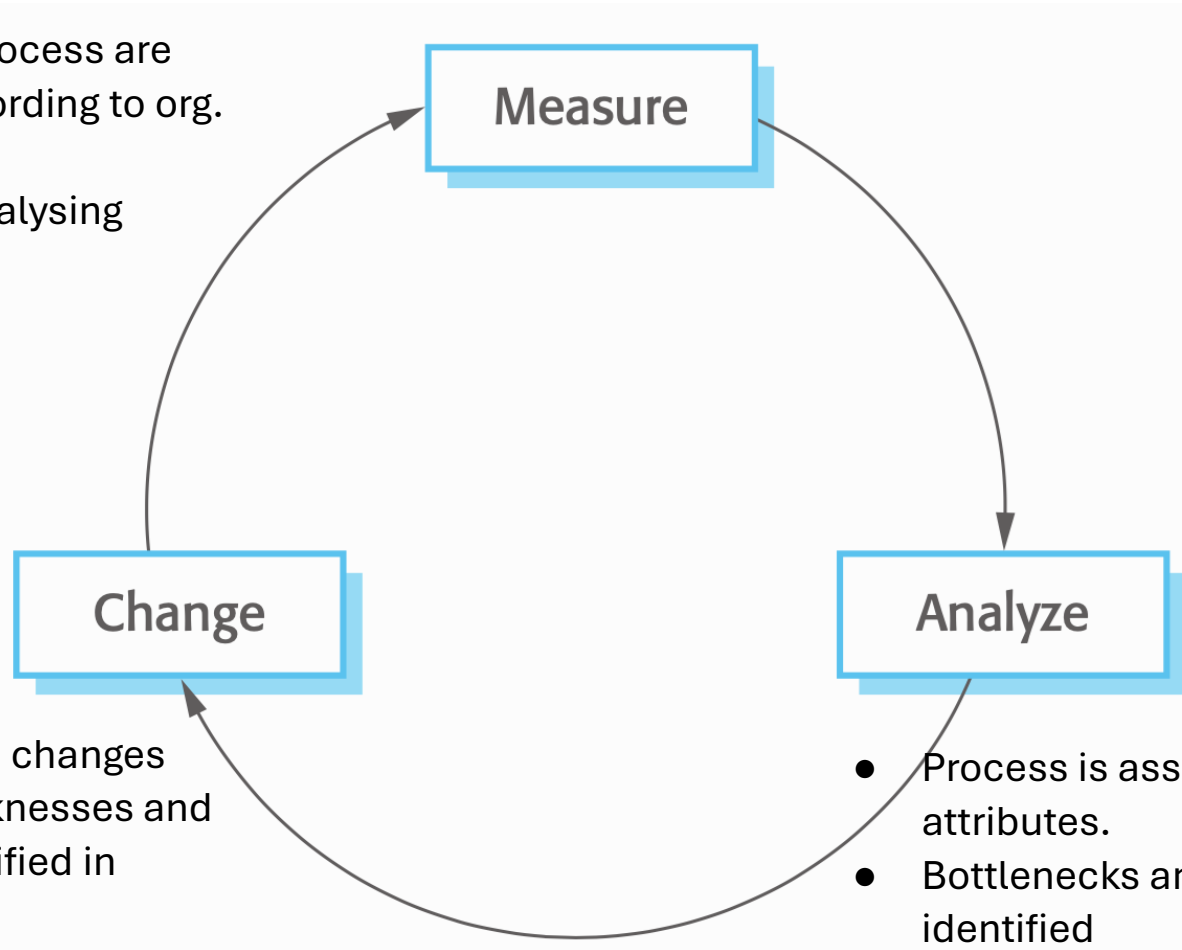
- Process improvement for years focused on defect reduction.
Nowadays, examples of process attributes are
 - Understandability:
 - How easy to understand the process definition?
 - To what extent is the software process define?
 - Standardization:
 - How is the process is based on generic standard processes? (clients require conformance)
 - Visibility:
 - Are the process activities have clear results and progress is externally visible?
 - Measurability:
 - Does the current process facilitates data collection so that process characteristics can be measured
 - Supportability:
 - Can the current process be supported by tools? To what extent?

Process attributes (cont.d)

- Acceptability:
 - Is the current process acceptable to and usable by software engineers?
- Reliability:
 - To what extent that the process takes into account error-free software product? (verification)
- Robustness:
 - Can the software process function regardless of unexpected problems?
- Maintainability:
 - Can the software process evolve and address changes of requirements for improvement?
- Rapidity:
 - How fast can the process of delivering the product from a given specification be completed?

Process Improvement: measure, analyze, change

- Attributes of process are measured according to org. goals
- Baseline for analysing improvements



- Process is assessed based on attributes.
- Bottlenecks and weaknesses are identified
- Rapidity and robustness are considered

Propose appropriate changes that addresses weaknesses and improvements identified in analysis stage

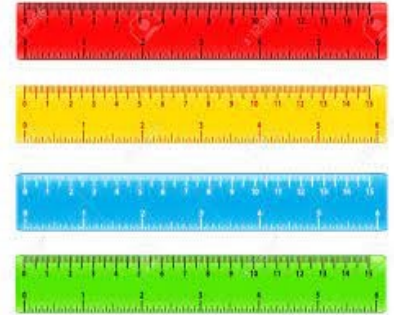
Process measurement

- Process measurements are quantitative data about software
 - E.g. time taken to perform a process to complete
 - Eg. time taken to develop test cases
 - Eg. time taken for the user to fill a electronic form
 - E.g. the effort devoted to do detailed model of system
- Also, product quality data must be collected and then related to process

Safety	Understandability	Portability
Security	Testability	Usability
Reliability	Adaptability	Reusability
Resilience	Modularity	Efficiency
Robustness	Complexity	Learnability

Process measurement (cont.d)

- Three types of process metrics can be collected:
 - Time taken for a particular process to be completed
 - Eg. time to complete modeling of the system
 - The resources required for a particular process to complete
 - Eg. total effort in person-days to do requirements gathering
 - The number of occurrences of a particular event
 - Eg. number of defects discovered during code inspection or pair programming
- Measurements cannot be used to drive improvements (only used for process analysis)
- Improvements should be driven by objectives



Process measurement (cont.d)

- The long-term goal of measurement is to review and make judgement about software/process quality
- Examples of metrics
 - Lines of code, number of reported faults, number of days required to develop a system feature
- Metrics may be either control metrics or predictor metrics
 - Control metric: support process management
 - Ex: average effort, time to repair faults, etc
 - Predictor metric: helps to predict characteristics of the software
 - Ex: number of attributes and operations of object class, average length of identifiers in a program,

Process measurement (cont.d)

- Sometimes it is difficult to know what information should be collected to support process improvement.
- Basili and Rombach proposed the “Goal-Question-Metric” paradigm
 - Goal: what organization is trying achieve (how to current process affects the products)
 - Eg. shorten the product time delivery
 - Questions: a goal must be associated with questions that need answers.
 - Eg. how can the time of developing the models can be reduced?
 - Metrics: theses are the measurements need to be collected to answer the questions.
 - Eg. how much time required to finish the modeling with current modeling tool?

Process analysis

- Study processes and help understand key characteristics
- Also, how processes are performed by the people involved
- Objectives:
 - To understand the activities involved in the process and their relationships
 - To understand the measurements that have been made from activities
 - To related processes to comparable processes elsewhere

Process analysis (cont.d)

- Aspects of process analysis
 - Adoption and standardization
 - Eg. is the process documented and followed across the organization?
 - Software engineering practice
 - Eg. are there good practices that are not included in this process?
 - Organization constraints
 - Eg. what constraints that affect the process that we followed?
 - Communication
 - Eg. to what extent communication issues contribute the project delay?
 - Learning
 - Eg. how new joining people learn about the software? Is there any process manuals?
 - Tool support
 - Eg. what parts of the process are and aren't supported by tools?

Process change

- Involves making modifications to existing software processes
- This may involve:
 - Introducing new practices, methods or processes
 - Changing the ordering of process activities
 - Introducing or removing deliverables
 - Introducing new roles or responsibilities
- Change should be driven by measurable goals

Process change challenges

There are two major challenges of change process stage

1. Resistance to change

- a. Introducing changes may be resisted by few people involved in the process
- b. They may deliberately go against process change and interpret data to show ineffectiveness changes

1. Change persistence

- a. Changes may discarded after short time and revert to previous state
 - i. Happen when the person who proposed changes is left to another firm

Process classification

- Informal
 - No detailed process model. Development team chose their own way of working
- Managed
 - Defined process model which drives the development process
- Methodical
 - Processes supported by some development method such as RUP
- Improving
 - Processes quantitatively understood analysed and changed to improve performance

Software standards

- Define attribute of a product or a process
- Play important role in quality management
- Standards can be
 - International,
 - National, (should be influenced by international norms of standards)
 - Organizational, (should be based on international and national ones)
 - Project standards (should be based on the organizational ones)

Software standards (cont.d)

- Standards are important for three reasons
 - They encapsulate best practices: evade repetition of past mistakes
 - They define what ‘quality’ means in a particular situation
 - They provide continuity: any individual joining the organisation can understand organizations’ standards easily.

Software standards (cont.d)

- Standards can be related to two types
 1. Product standards: related to the software being developed.

Ex: documents standards (IEEE documentation), coding standards
 2. Process standards: define the processes that should be followed during development.

Ex: definition of specification, design and validation process

Software standards (cont.d)

- Problems with standards:
 - They may be seen by engineers as overprescriptive or not up-to-date
 - They often involve too much bureaucratic form filling
 - Some may not be supported by tools
 - Very hard and time consuming then to be done and validated manually

Software standards (cont.d)

- ISO 9001 standards framework
 - International set of standards for developing quality systems
 - Including software
 - Sets out quality principles, describes quality processes
 - Also, lays out the organizational standards and procedures that should be defined and documented

Software standards (cont.d)

- ISO 9001 and quality management: for organization to be conformant it must follow adapt the following structure

